

JCL Lab 1,

Chapter 6 Exercise

Part 1

This lab exercise helps you develop skills in creating batch jobs and submitting them for execution on z/OS.

NOTE: Copy the member JCLTEST from the data set 'CWSEAY.JCL' TO YOUR LANG.CNTL DATASET using the 3.3 move/copy function. Make sure you use single quotes since is MY dataset. You have read-only access to CWSEAY.JCL. Note the single quotes for the data set. This is because you are accessing it, but it is not your data set. We will copy other files from this library later in the course.

```
Marist
File Edit View Tools Help
F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12

Menu RefList Utilities Help

Move/Copy Utility

C Copy data set or member(s)          CP Copy and print
M Move data set or member(s)         MP Move and print

Specify "From" Data Set below, then press Enter key

From ISPF Library:
Project . . . . . (--- Options C and CP only ---)
Group . . . . .
Type . . . . .
Member . . . . . (Blank or pattern for member list,
                  "*" for all members)

From Other Partitioned or Sequential Data Set:
Name . . . . . 'KC02071.ECUJCL(JCLTEST)'
Volume Serial . . . . . (If not cataloged)
Data Set Password . . . . . (If password protected)

Option ==> C
*MCOPI
```

```
Marist
File Edit View Tools Help
F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 PA1 Attn Sysreq
Menu RefList Utilities Help
-----
COPY      From KC02071.ECUJCL(JCLTEST)

Specify "To" Data Set Below

To ISPF Library:                                Options:
Project . . .                                     Enter "/" to select option
Group . . .                                     Replace like-named members
Type . . .                                     Process member aliases
Member . . .                                     (Blank unless member is to be renamed)

To Other Partitioned or Sequential Data Set:
Name . . . . . LANG.CNTL
Volume Serial . . . (If not cataloged)

Data Set Password . . (If password protected)

To Data Set Options:
Sequential Disposition      Pack Option      SCLM Setting
1 1. Mod                    3 1. Yes          3 1. SCLM
2 2. Old                    2 2. No           2 2. Non-SCLM
3 3. Default                3 3. As is        3 3. As is

Command ==>
F1=Help      F2=Split      F3=Exit      F7=Backward  F8=Forward  F9=Swap
F10=Actions  F12=Cancel
```

```
Marist
File Edit View Tools Help
F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 PA1 Attn Sysreq
Menu RefList Utilities Help
-----
Move/Copy Utility                                Member JCLTEST copied

C Copy data set or member(s)                    CP Copy and print
M Move data set or member(s)                    MP Move and print

Specify "From" Data Set below, then press Enter key

From ISPF Library:
Project . . . (--- Options C and CP only ---)
Group . . .
Type . . .
Member . . . (Blank or pattern for member list,
              "*" for all members)

From Other Partitioned or Sequential Data Set:
Name . . . . . KC02071.ECUJCL(JCLTEST)
Volume Serial . . . (If not cataloged)

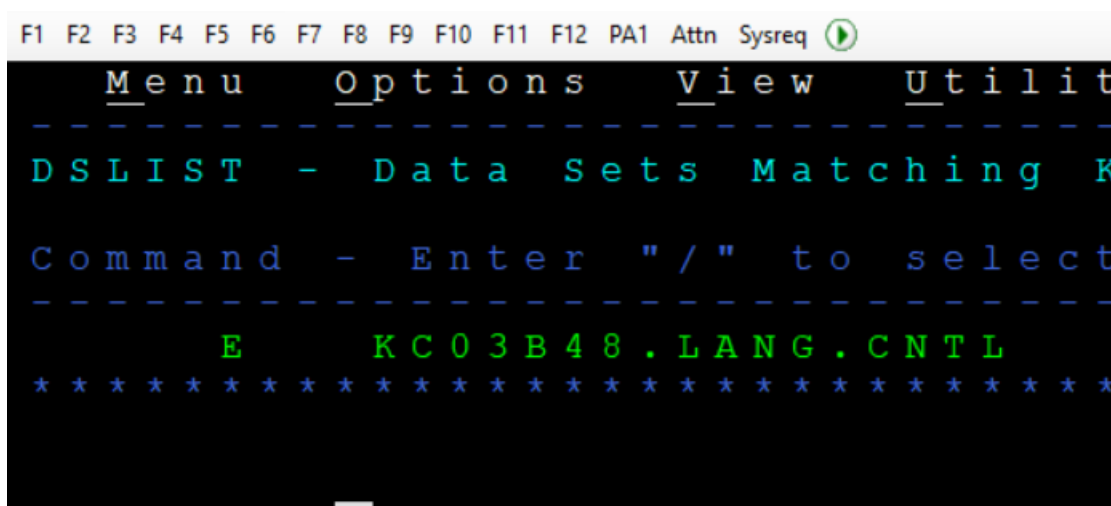
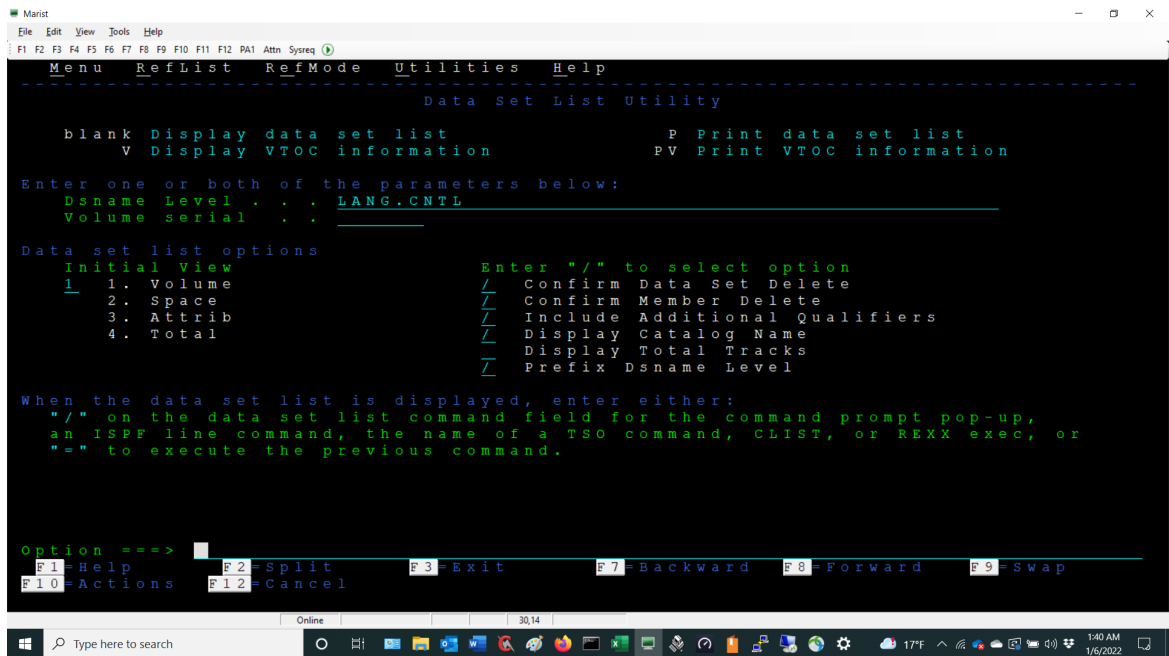
Data Set Password . . (If password protected)

Option ==> C
F1=Help      F2=Split      F3=Exit      F7=Backward  F8=Forward  F9=Swap
F10=Actions  F12=Cancel
```

The data set JCLTEST is now in YOUR LANG.CNTL library (NOT ECUJCL, that's for another class)

Part 2 Creating a simple job

From ISPF, navigate to the Data Set List Utility (Option 3.4) panel and enter **LANG.CNTL** in the Dsname Level field(described in an earlier exercise) and press ENTER. **MAKE SURE THERE IS A SLASH IN THE PREFIX DSNAME LEVEL FIELD**



```

Name
-----
CALC1000
CALC2MOD
CALC2000
CALC3000
CHAR2200
DATE1100
DATE2200
ECUSORT
HELLO
HELLO1
JCLTEST

```

Notice that only a single JCL line is in the data set, **//STEP1 EXEC PGM=IEFBR14**, you are going to enter this job, but understand it does not have a JOB statement.

```

EDIT          KC03A99.ECUJCL(JCLTEST)
*****
==MSG> -Warning- The UNDO command
==MSG>          your edit profile
000001 //STEP1 EXEC PGM=IEFBR14
*****

```

Take a screenshot of this screen and label it "My job without a Job name."

- To submit the JOB, you type SUB in the command line as below. Since there is no JOB statement at this point, the system will give your job a name by default. The default jobname is to add a character to your ID. Enter any character or number in response to the message shown:

```

EDIT          KC03A99.ECUJCL(JCLTEST) - 01.32          Columns 00
*****          ***** Top of Data *****
==MSG> -Warning- The UNDO command is not available until you change
==MSG>          your edit profile using the command RECOVERY ON.
000001 //STEP1 EXEC PGM=IEFBR14
*****          ***** Bottom of Data *****

Command ==> sub
*DSLST

ENTER JOBNAME CHARACTER(S) -
A

ENTER JOBNAME CHARACTER(S) -
A

JOB KC03A99A(JOB00544) SUBMITTED
***

```

The result will be the message:

JOB [YOURIDA](JOB[some job number]) SUBMITTED

Print Screen (PS) THE MESSAGE THAT INDICATES YOUR JOB HAS BEEN SUBMITTED AND LABEL IT "MY JOB HAS BEEN SUBMITTED". Note that because you did not have a job statement the system used YOURID plus a character to make an 8 character job name.

Note: Whenever you see three asterisks (***) , it means there's more data to see. Press Enter to continue.

When the job finishes, you should see the message:

\$HASP165 yourid1 ENDED AT SYS1 MAXCC=0000 CN(INTERNAL)

In the file/JOB JCLTEST add (insert) a new first line in your file that will hold a JOB statement. The JOB statement must precede the EXEC statement. After you insert a line, you can move the EXEC statement down one line as follows:

```
=MSG>                                your edit profile
001001  //STEP1 EXEC PGM=IEFBR14
*****
```

Place an "m" (for move) to the left of the EXEC statement, and an "a" (for move after) on the line below it:

```
=MSG>                                your edit profile
000m01  //STEP1 EXEC PGM=IEFBR14
'a'
*****
```

Press ENTER, and the EXEC statement will be moved a line down:

```
*****
000001  //STEP1 EXEC PGM=IEFBR14
*****
```

Then enter a JOB statement named [YOURID]A (your ID with the letter A at the end)

```
*****
000001  //KC03A99A JOB 1
000002  //STEP1 EXEC PGM=IEFBR14
*****
```

At this point, the member JCLTEST should read similar to this (with your ID instead of the one shown). Please note that the use of your ID is not system defined. It is just used as the default job name. You can use any set of characters 8 or less:

```
//KC03A99A JOB 1
//STEP1 EXEC PGM=IEFBR14
```

PS THIS SCREEN WITH YOUR ADDED LINE. Label it "JOB STATEMENT ADDED"

Type **SAVE** at the command line and then **SUB** to submit your job. You will notice that you are not prompted to add a character this time, and the Job has a name [YOURID]A, and a Job

number in parenthesis.

```
***** Top of Data
000001 //KC03A99A JOB 1
000002 //STEP1 EXEC PGM=IEFBR14
***** Bottom of Data

Command ==> sub
          *DSLIST

JOB KC03A99A (JOB00637) SUBMITTED
***
```

Exit ISPF.

From the ISPF Primary Option Menu, find SDSF (Option SD on our system).

You can toggle two screens by pressing F2 twice quickly (invoking the split screen) and using F9 to toggle between screens. Practice this a couple of times to get used to it. Using two screens will come in handy in the next several exercises. We will show you in class a way to toggle more than two screens.

In the SDSF menu, enter **PREFIX *** from the SDSF command line, then enter **OWNER [YOURID]** from the command line (Ex. OWNER KC03A38). This will make sure you get all of your jobs and no one else's.

```
PARM Parmlib data sets
PROC Proclib data sets
COMMAND INPUT ==> PREFIX *_
*SDSF
```

```
PROC Proclib data sets
COMMAND INPUT ==> OWNER KC03G1F_
*SDSF
```

Please make sure you type PREFIX * and not PREFIX [YOURID]. Students sometimes make that mistake. They also sometimes type OWNER * and that leads to a mess.

Then enter **ST** (Status Panel). Both jobs that you submitted should be listed. Place **S** (select) to the left of either job, press **ENTER**, then page up and down (**F8**- pg down, **F7** – pg up) to view the messages produced from the execution.

```
COMMAND INPUT ==> ST _
*SDSF
```

```

      KC03G1F  TSU09599 KC03G1F  15 EXECUTION
      KC03G1F  TSU09452 KC03G1F   1 PRINT
S_    KC03G1FA JOB09628 KC03G1F   1 PRINT      A

```

```

Vista TN3270 Session 8
File Edit Font Transfer Macro Options Window Help
[Icons] [A B C ?]

Display Filter View Print Options Search Help
-----
SDSF OUTPUT DISPLAY KC03N771 JOB09456 DSID 2 LINE 0 COLUMNS 02- 81
COMMAND INPUT ==> _ SCROLL ==> PAGE
***** TOP OF DATA *****
J E S 2 J O B L O G -- S Y S T E M S O W 1 -- N O D E

02.02.02 JOB09456 ---- TUESDAY, 24 SEP 2013 ----
02.02.02 JOB09456 IRR010I USERID KC03N77 IS ASSIGNED TO THIS JOB.
02.02.02 JOB09456 ICH70001I KC03N77 LAST ACCESS AT 01:59:18 ON TUESDAY, SEPT
02.02.02 JOB09456 $HASP373 KC03N771 STARTED - INIT 2 - CLASS A - SYS SOW1
02.02.02 JOB09456 - ----TIMINGS (MINS.)--
02.02.02 JOB09456 -STEPNAME PROCSTEP RC EXCP CONN TCB SRB C
02.02.02 JOB09456 -STEP1 00 7 1 .00 .00
02.02.02 JOB09456 -KC03N771 ENDED. NAME-KC03N77 TOTAL TCB CPU TIM
02.02.02 JOB09456 $HASP395 KC03N771 ENDED

----- JES2 JOB STATISTICS -----
24 SEP 2013 JOB EXECUTION DATE
5 CARDS READ
32 SYSOUT PRINT RECORDS
0 SYSOUT PUNCH RECORDS
F1=HELP F2=SPLIT F3=END F4=RETURN F5=IFIND F6=BOOK
F7=UP F8=DOWN F9=SWAP F10=LEFT F11=RIGHT F12=RETRIEVE

MA 0.0 09/24/13.267 03:15AM zos.kctr.marist.edu A 5,21

```

PS THE THIS PAGE OF YOUR DISPLAYED JOB, label it “OUTPUT OF MY JOB” THEN PRESS F3 TO EXIT.

Edit JCLTEST again (you should be able to f9 to switch screens), and insert the following lines at the bottom of the Job (using i[x]- where the x = some number of lines to be inserted):

```
//CREATE DD DSN=yourid.MYTEST,DISP=(NEW,CATLG),
// UNIT=SYSDA,SPACE=(TRK,1)
```

```

EDIT          KC03B48.LANG.CNTL(JCLT
*****
==MSG> -Warning- The UNDO command
==MSG> your edit profil
000001 //KC03A48 JOB 1
00i402 //STEP1 EXEC PGM=IEFBR14
*****

```

```

EDIT          KC03B48.LANG.CNTL(JCLTE
*****
==MSG> -Warning- The UNDO command
==MSG> your edit profile
000001 //KC03A48 JOB 1
000002 //STEP1 EXEC PGM=IEFBR14
*****

```

```

EDIT          KC03B48.LANG.CNTL(JCLTEST) - 01.33
***** ***** Top of Data *****
==MSG> -Warning- The UNDO command is not available until
==MSG> your edit profile using the command R
000001 //KC03A48 JOB 1
000002 //STEP1 EXEC PGM=IEFBR14
***** //CREATE DD DSN=yourid.MYTEST,DISP=(NEW,CATLG),
***** // UNIT=SYSDA,SPACE=(TRK,1)
***** ***** Bottom of Data *****

```

What these lines are doing is using JCL to create a data set. The name of the data set is YOURID.MYTEST, the DISP disposition means that it is a new data set and will be cataloged with a normal end. There is no contingency for an abnormal end but the system default is to delete the file on an abnormal end. The second line says use a system volume and use tracks as how you organize the records. There are system defaults for everything else needed to store the data set.

MAKE SURE TO REPLACE YOURID with YOUR ID NUMBER!

[NOTE THE SPACE BETWEEN // AND UNIT FOR THE CONTINUATION OF THE PREVIOUS LINE [NOTE THE COMMA AT THE END OF THE LINE]; THIS IS A COMMON ERROR MADE BY ALL (INCLUDING ME)]

PS THIS SCREEN WITH THE LINES YOU ADDED TO THE JOB, AND LABEL IT "I AM CREATING A NEW DATASET NAME [yourid].MYTEST; IT WILL BE CATALOGED).

Submit the content of JCLTEST created above, press PF3 (save and exit edit), then view the output of this job using SDSF (using the F9 toggle). Notice that you have more than one job with the same jobname. The jobname with the highest JOBID number is the last one that was run.

What was the condition code? If it is something other than 0, page down to the bottom of the output listing to locate the JCL error message. Correct JCLTEST and resubmit. Ask for help if you cannot find the problem. Repeat until cond code=0000 is received. A Condition Code 000 means the job ran with no errors.. Navigate to the Data Set List Utility panel (=3.4) and enter *yourid*.MYTEST in the DSNAME level field. What volume was used to store the data set?

PS THE SCREEN THAT SHOWS YOU CREATED A NEW DATASET

IEFBR14 IS USEFUL TO RUN A JOB TO CREATE A DATA SET, BECAUSE IT DOES NO I/O. IN CASE YOU DIDN'T NOTICE, WHAT YOU DID WAS USE A JOB TO CREAE A SEQUENTIAL DATASET.

Finally, write 50-150 words describing what you have done in this lab and turn this in with your screen shots.